

Metallicity Histories in AuroraLF

Representative halos ending at $z = 12.5$

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Question

- Track enrichment along Monte Carlo MAHs.
- Compare canonical, redshift-gated, and burst-gated top-heavy modes.
- Keep UV/SFR physics fixed; use birth metallicity in the IMF gate.

Representative grid

$$\begin{aligned} z_{\text{final}} &= 12.5, \\ \log_{10} M_h(z_{\text{final}}) &= 9, 10, 11, 12, \\ N_{\text{track}} &= 1000, \quad N_z = 240. \end{aligned}$$

Histories show median and 16–84 percentile ranges across valid tracks.

One-Zone Metal Reservoir

Reservoir and birth metallicity

$$M_{\text{gas},i} = f_{\text{gas}} f_b M_{h,i},$$

$$Z_{\text{gas},i} = \frac{M_{Z,i}}{M_{\text{gas},i}},$$

$$Z_{\text{birth},i} = \frac{M_{Z,i^-}}{M_{\text{gas},i} Z_{\odot}} 10^{\mathcal{N}(0, \sigma_{\text{birth}})}.$$

Effective yield

$$y_i = y [1 + (m_{\text{TH}} - 1) l_{\text{TH},i}] 10^{\mathcal{N}(0, \sigma_y)},$$

$$y = 0.02, \quad m_{\text{TH}} = 1.28.$$

$$\Delta M_{Z,i} = y_i \Delta M_{\star,i} + Z_{\text{in}} \Delta M_{\text{in},i} - Z_{\text{gas},i-1} [(1 - R) + \eta_i] \Delta M_{\star,i}.$$

Top-Heavy Source-Time Gates

IMF flag

$$I_{\text{TH}} = 0$$

canonical,

$$I_Z = \mathbf{1}[Z_{\text{birth}} \leq 0.05 Z_{\odot}],$$

$$I_{\text{TH}}^{10} = \mathbf{1}[z_{\text{src}} \geq 10] I_Z,$$

$$t_{\text{grow}} = M_h / \dot{M}_h,$$

$$I_{\text{TH}}^{\text{burst}} = I_{\text{TH}}^{10} \mathbf{1}[t_{\text{grow}} \leq 50 \text{ Myr}].$$

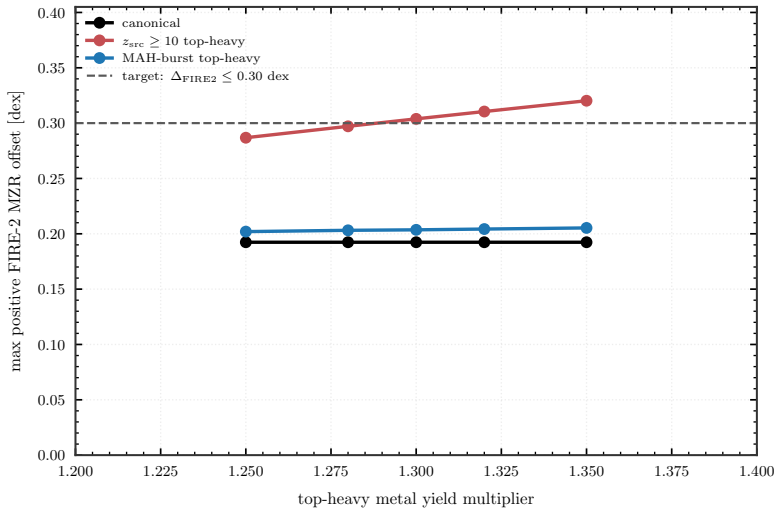
Displayed summaries

median $[Z_{\text{gas}}/Z_{\odot}]$ over active halos,

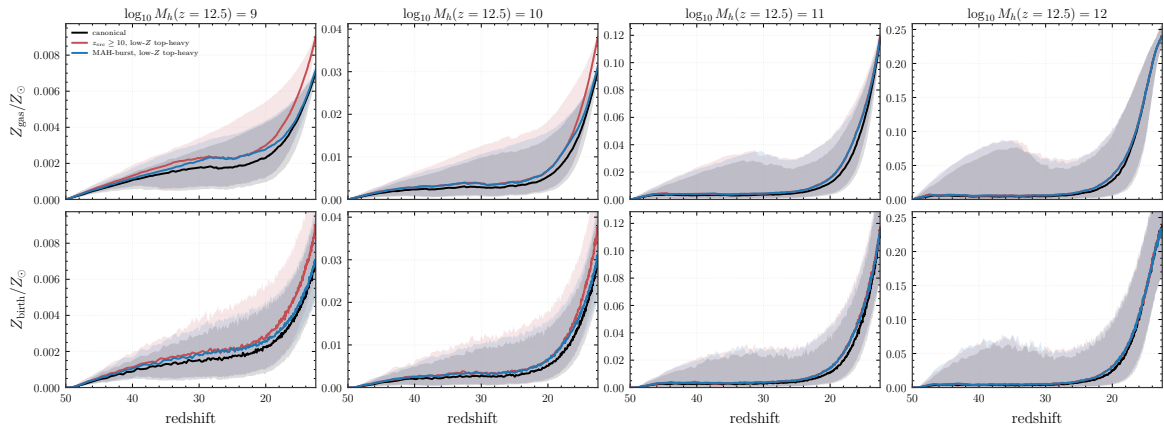
median $[Z_{\text{birth}}/Z_{\odot}]$ over star-forming cells.

- Bands: 16–84 percentile range.
- Same MAH seed per mass for all modes.

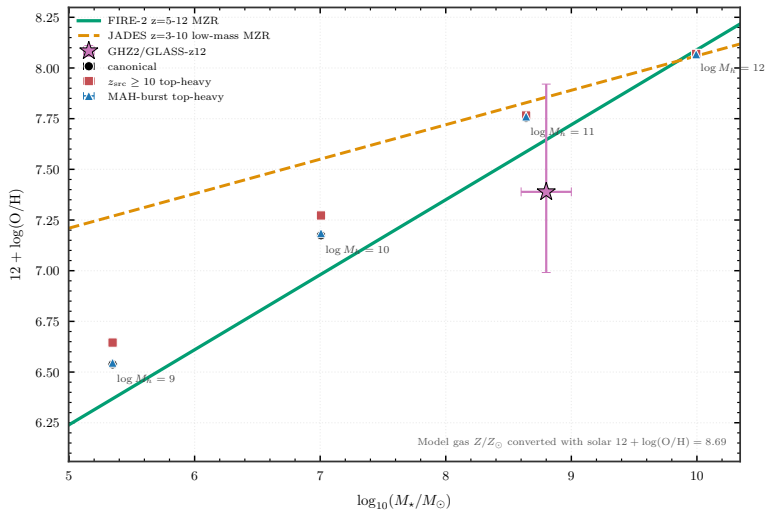
Yield Multiplier Calibration



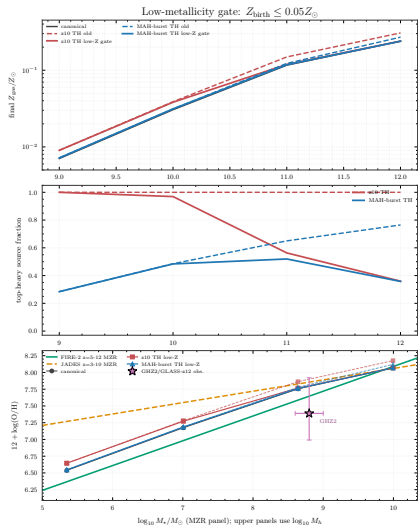
Metallicity Histories



Literature MZR Check



Effect of the Low-Metallicity Gate



Final Gas Metallicity at $z = 12.5$

$\log_{10} M_h$	canonical	$z_{\text{src}} \geq 10$ TH	MAH-burst TH	$f_{\text{TH,src}}$ (MAH)
9	0.00705	0.00902	0.00719	0.284
10	0.0305	0.0389	0.0313	0.485
11	0.116	0.119	0.118	0.519
12	0.239	0.239	0.239	0.358

Mass trend

- Final enrichment rises strongly with final halo mass.
- High-mass MAHs enter the burst gate more often.

Mode trend

- z_{10} is strongly suppressed once $Z_{\text{birth}} > 0.05Z_{\odot}$.
- MAH-burst remains close to canonical at all four masses.

- **Low-Z gate:** top-heavy requires $Z_{\text{birth}} \leq 0.05Z_{\odot}$.
- **High-mass halos:** z10 top-heavy source fraction falls from 1.00 to 0.36 at $\log M_h = 12$.
- **MZR effect:** maximum positive FIRE-2 offset changes from +0.297 to +0.290 dex for z10.
- **JADES effect:** maximum positive JADES excess drops from +0.117 to +0.010 dex.
- SFR and dust remain fixed; only SSP choice and metal yield use the new IMF gate.

References

-  Reed et al. 2007, halo mass function fitting functions.
-  Curti et al. 2023, JADES low-mass MZR at $3 < z < 10$.
-  Marszewski et al. 2024, FIRE-2 gas-phase MZR at $z = 5\text{--}12$.
-  Castellano et al. 2024, GHZ2/GLASS-z12 metallicity at $z = 12.33$.
-  Eldridge et al. 2017, BPASS stellar population synthesis.
-  Stanway & Eldridge 2018, BPASS v2.2 binary populations.
-  AuroraLF local implementation, `codex/topheavy-stochastic-metallicity`.

Metallicity histories are now available as per-track diagnostics for the top-heavy IMF branches.

Backup: Output Files

Product	Prefix
history outputs	metallicity_history_z12p5_mth1p28_lowzgate_20260519
slide asset	slides/assets/metallicity_history_z12p5.pdf
MZR comparison	metallicity_mzr_literature_z12p5_mth1p28_lowzgate_20260519
old/new comparison	metallicity_lowzgate_comparison_z12p5_mth1p28_20260519

Backup: Birth Metallicity at $z = 12.5$

$\log_{10} M_h$	canonical	$z_{\text{src}} \geq 10$ TH	MAH-burst TH
9	0.00701	0.00889	0.00692
10	0.0298	0.0380	0.0312
11	0.117	0.117	0.115
12	0.238	0.233	0.234

Birth metallicity is sampled only for star-forming source-time cells.